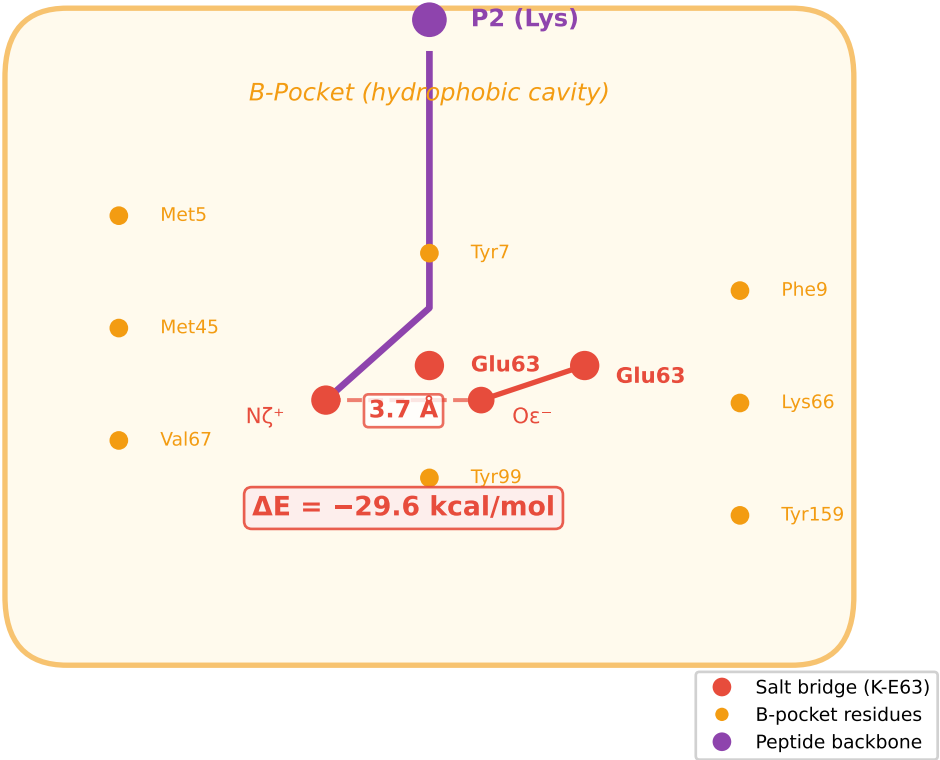
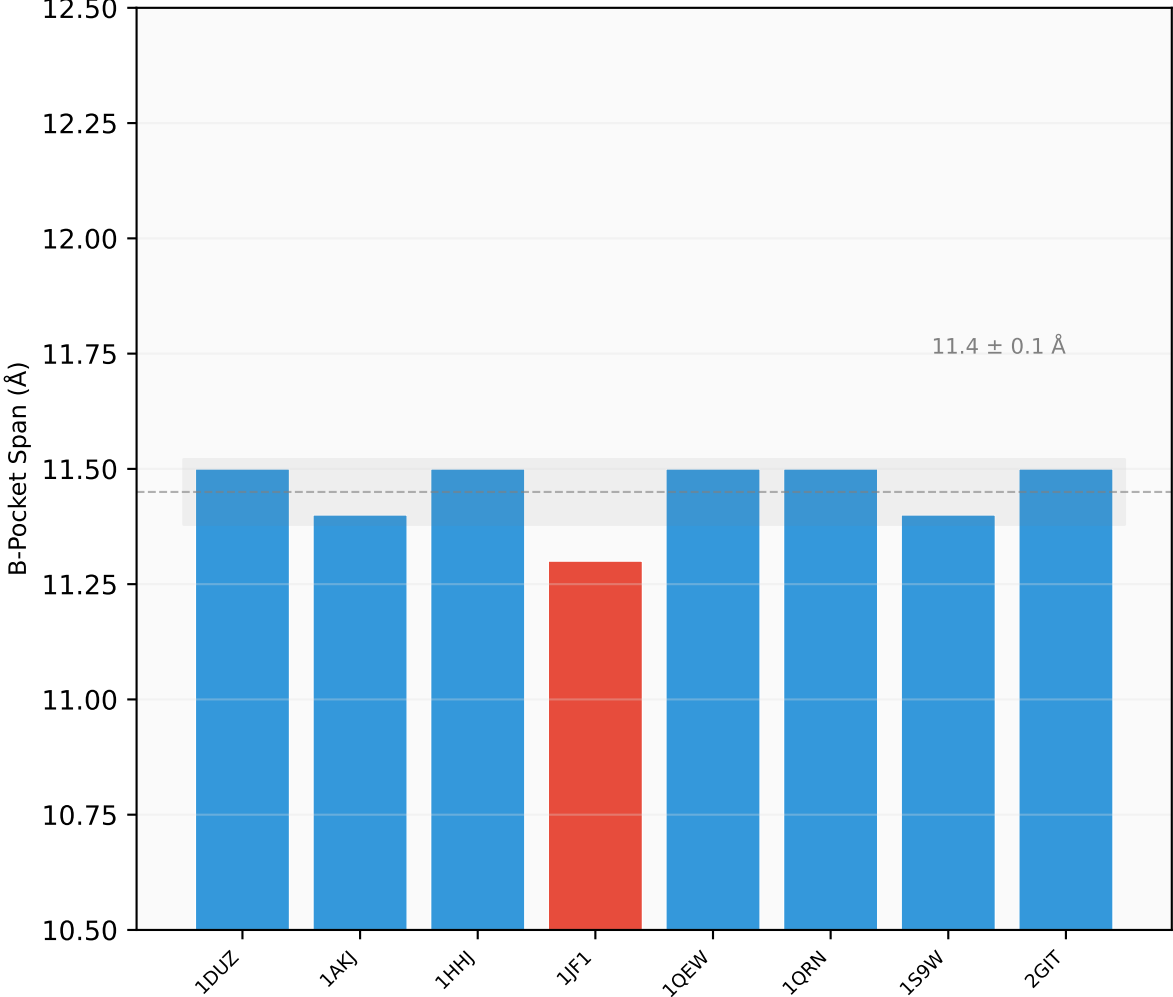


Structural Basis of Non-Canonical Peptide-MHC Anchoring
KRAS G12V Neoepitope (YKLVVVGAV) · HLA-A*02:01 · K-E63 Salt Bridge

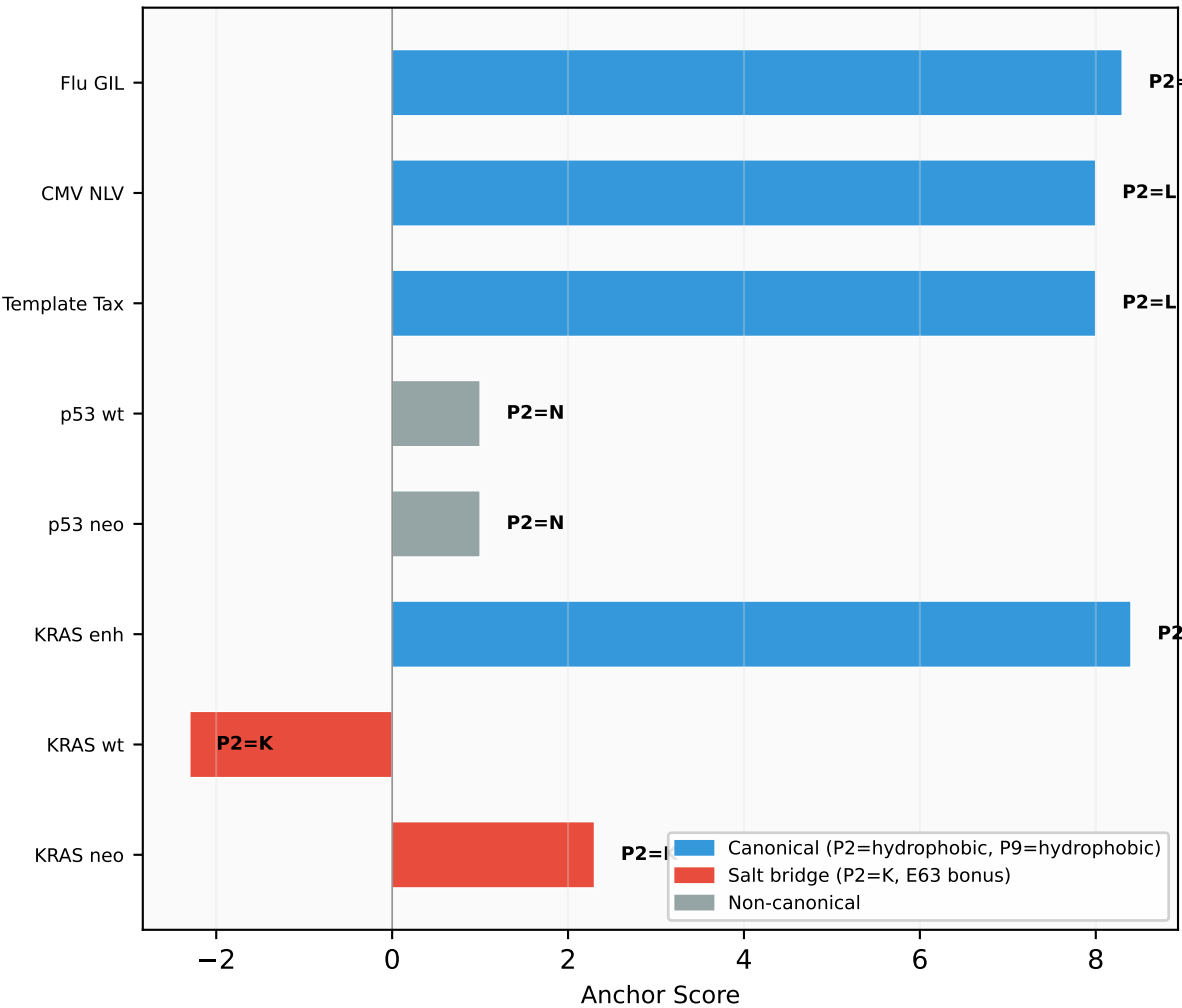
A B-Pocket Salt Bridge: K(P2) → Glu63



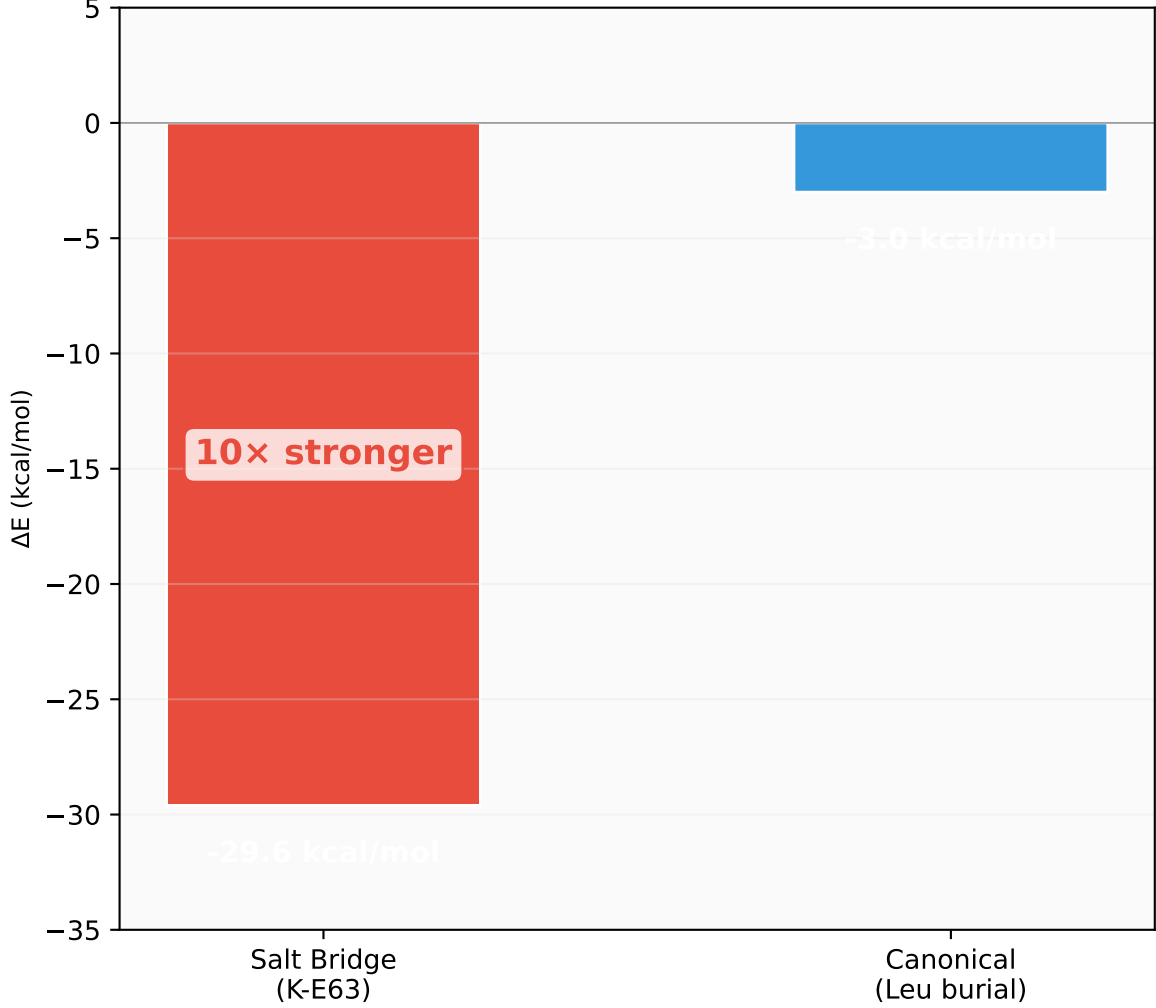
B B-Pocket Conservation (8 HLA-A*02:01 Structures)



C Peptide Anchor Scores (P2 + P9 Motif)



D Binding Energy: Salt Bridge vs Canonical Anchor



E K-E63 Salt Bridge — Structural Evidence Summary

Method:

Template:

P2 CA → Glu63 OE2:

P2 CA → Glu63 OE1:

Min OE distance:

K sidechain reach (CA→Nz):

Salt bridge cutoff:

Coulomb ΔE (salt bridge):

Canonical Leu burial ΔG:

Salt bridge advantage:

B-pocket conservation (8 PDBs):

Verdict:

pdffixer + OpenMM minimization (amber14, no cutoff)

1DUZ (HLA-A*02:01 + LLFGYPVYV, 1.80 Å)

3.7 Å

3.9 Å

3.7 Å

7.6 Å

4.0 Å (standard N-O definition)

-29.6 kcal/mol

-3.0 kcal/mol (hydrophobic anchor)

~10x vs canonical

11.4 ± 0.1 Å

PLAUSIBLE — structurally & energetically favorable

First documented charged P2 anchor

in HLA-A*02:01 binding